



SUPPLY CHAIN

& INVENTORY ANALYTICS

Assignment





Dataset Link: [Supply_Chain_dataset](#)

Hint Visual Requirements:

Question	Visual(s) to Create
Q1	No Visual (Data Import & Validation)
Q2	No Visual (Data Modeling)
Q3	4 KPI Cards
Q4	Clustered Column Chart, Clustered Bar Chart, Treemap
Q5	Donut Chart, Clustered Bar Chart, Clustered Column Chart
Q6	Clustered Bar Chart, Matrix
Q7	Funnel Chart, Table
Q8	Line & Clustered Column (Combo) Chart, Clustered Column Chart
Q9	Clustered Bar Chart, Clustered Column Chart, Donut Chart
Q10	Clustered Bar Chart, Clustered Column Chart, Donut Chart
Q11	Clustered Column Chart, Clustered Bar Chart, Donut Chart
Q12	Drill-through Page (Reuse 3 KPI Cards, 1 Table, Back Button)
Q13	Executive Dashboard (Reuse 4 KPI Cards, 4 Business Visuals, 3 Dropdown Slicers)

Submission Guidelines

- Complete all 13 questions in the given sequence.
- Save the Power BI report as Supply Chain_Inventory_Analytics.pbix.
- Ensure all relationships, measures, and visuals are working correctly.
- Apply appropriate titles and formatting to all report pages.
- Verify that all slicers, Drill-through, and navigation features function correctly.
- Include the Executive Dashboard and Product Details (Drill-through) pages.
- Submit the .pbix file along with screenshots of the Executive Dashboard and Drill-through page (if required).
- Ensure the report is free from errors before submission.

Q1. Data Import and Data Validation

Question

Import all six CSV files into Power BI Desktop and perform data validation using Power Query.

Perform the following tasks:

- Import all six CSV files successfully.
- Verify and correct the data type of each column.
- Identify and handle missing values (if any).
- Check for duplicate records and remove them where necessary.
- Ensure all tables are clean and ready for analysis.

Deliverables:

- Screenshot of the imported tables.
- Screenshot of the Power Query Editor after data validation.

- Summary of the data quality checks performed.
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Q2. Data Modeling

Question

Create an optimized data model using the imported tables.

Your model should:

- Identify the Fact table and Dimension tables.
- Create appropriate relationships between the tables.
- Configure the correct cardinality and cross-filter direction.
- Ensure all dimension tables correctly filter the fact table using a Star Schema.

Deliverables:

- Screenshot of the Model View.
 - Brief explanation of the relationship design.
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Q3. Create Business Measures

Question

Create a separate **Measures** table and develop the following DAX measures:

- Total Sales
- Total Orders
- Total Quantity Sold
- Average Order Value

Display all measures using KPI Cards with appropriate formatting.

Q4. Product Performance Analysis

Question

Develop a report to analyze product performance.

Include the following analyses:

- Total Sales by Product Category
- Total Sales by Brand
- Top 10 Products based on Total Sales

Write **three business insights** based on your analysis.

Q5. Customer Segment Analysis

Question

Develop a report to analyze customer purchasing behavior across different customer segments.

Include the following analyses:

- Total Sales by Customer Segment
- Total Orders by Customer Segment
- Total Sales by Region

Write **three business insights** based on your analysis.

Q6. Warehouse Performance Analysis

Question

Develop a report to evaluate the performance of different warehouses.

Include the following analyses:

- Total Sales by Warehouse
- Total Quantity Sold by Warehouse
- Average Delivery Days by Warehouse
- Create **Matrix Card of Warehouse name, Total Quantity Sold, Average Delivery Days**

Write **three business insights** based on your analysis.

Q7. Supplier Performance Analysis

Question

Create a report to analyze supplier performance.

Include the following analyses:

- Total Sales by Supplier (Funnel Chart)
- Total Quantity Sold by Supplier
- Top 10 Suppliers based on Total Sales
- Create Table Visual of Top 10 Supplier Name , Total Quantity & Total Sales

Write **three business insights** based on your analysis.

Q8. Monthly Sales Trend Analysis

Question

Develop a report to analyze monthly sales performance and product category sales.

Create the following visuals:

- Monthly Total Quantity Sold and Total Sales.
- Total Sales by Product Category.

Write **three business insights** based on your analysis.

Q9. Discount Analysis

Question

Develop a report to analyze the impact of discounts on sales across different product categories.

Create the following visuals:

- Average Discount (%) by Product Category.
- Total Sales by Discount Percentage.
- Average Discount (%) by Supplier Type.

Write **three business insights** based on your analysis.

Q10. Delivery Performance Analysis

Question

Develop a report to evaluate delivery performance across warehouses and suppliers.

Create the following visuals:

- Average Delivery Days by Warehouse.
- Average Delivery Days by Supplier Type.
- Distribution of Orders by Delivery Status.

Write **three business insights** based on your analysis.

Q11. Return Analysis

Question

Develop a report to analyze product returns and identify areas requiring improvement.

Create the following visuals:

- Total Returned Orders by Product Category.
- Total Returned Orders by Warehouse.
- Return Rate (%) by Supplier Type.

Write **three business insights** based on your analysis.

Q12. Drill-through & Interactive Navigation

Question

Implement **Drill-through** functionality to enable users to navigate from a summary report to a detailed product report.

Perform the following tasks:

- Create a new report page named **Product Details**.
 - Configure **Drill-through** using **Product Name**.
 - Copy the required KPI Cards and Table visual from previous report pages.
 - Add a **Back Button** to return to the previous page.
 - Verify that the page displays details only for the selected product.
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Q13. Executive Dashboard

Question

Create a professional **Executive Dashboard** by combining the visuals created in the previous questions. Instead of creating new visuals, **copy and paste the existing visuals** onto a new report page and arrange them into a well-structured dashboard.

The dashboard must include:

- 4 KPI Cards
- 4 Business Visuals
- 3 Slicers

Apply proper alignment, formatting, and titles to create an interactive dashboard.